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PERFORMANCE SPECIFICATION

FLANGES, PIPE, BRONZE (SILVER BRAZING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 50-, 100-, 150-, 250-, and 400-pound silver brazing bronze pipe flanges for use with water, oil, gas, or steam service not to exceed 425 °F.

1.2 Classification. The flanges are of the following types, classes, sizes, and ratings, as specified (see 6.2).

1.2.1 Types. Flanges are of the following types:

a. Type PR – With groove for preinserted rings of silver brazing alloy

b. Type EF – Without preinserted rings (end or face feed type)

1.2.2 Classes, sizes, and ratings. Flanges are of the classes, sizes, and ratings as specified in table I.

TABLE I. Classes, sizes, and ratings.

Classes	NPS (nominal pipe size) (inches)	ASME ratings [1/]	Applicable tables
Plain:			
50 pounds	14 to 40, inclusive	50 lb/in^2 at 150 °F	VI, X, XII
150 pounds	¼ to 12, inclusive	150 lb/in^2 at 150 °F 100 lb/in^2 at 425 °F	VII, XI, XIII XIII
250 pounds	¼ to 12, inclusive	250 lb/in^2 at 150 °F 150 lb/in^2 at 450 °F	VIII, XI XIV
400 pounds	¼ to 10, inclusive	400 lb/in^2 at 150 °F 200 lb/in^2 at 425 °F	IX, XI XV
Brazing rings:			
50 pounds	14 to 40, inclusive	--	XVI
100, 150, 250, and 400 pounds	¼ to 12, inclusive	--	XVII

NOTE:
[1/] ASME rating refers to the pressure-temperature ratings of ASME B16.5 and ASME B16.34.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.3 Part or identifying number (PIN). PINs for flanges covered by this specification are constructed as shown in table II:

TABLE II. PIN construction.

M	20042	-	XX	XXX	-	XXXX
Prefix for Military Specification	Specification Number	-	Type PR or EF (see 1.2.1)	Class 050, 100, 150, 250, or 400 (see 1.2.2)	-	NPS (see 1.3.1)
Example PINs: a. M20042-PR150-0600 is an NPS 6, class 150-flange with preinserted groove and brazing ring. b. M20024-EF050-1200 is an NPS 12, class 50-flange without preinserted brazing ring.						

1.3.1 NPS PIN code. The NPS PIN code is a four-digit digital representation of NPSs from ¼ to 40. For example, NPS 2½ is PIN code "0250", NPS 5½ is PIN code "0550", and NPS 38 is PIN code "3800".

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication, Welding and Inspection, Casting Inspection and Repair for Machinery, Piping and Pressure Vessels

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the TMIN without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.34 - Valves Flanged, Threaded and Welding End

ASME B16.5 - Pipe Flanges and Flanged Fittings: NPS ½ through NPS 24 Metric/Inch Standard

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ASME B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

(Copies of these documents are available online at www.asme.org.)

AMERICAN WELDING SOCIETY (AWS)

AWS A5.8M/A5.8 - Specification for Filler Metals for Brazing and Braze Welding

(Copies of this document are available online at www.aws.org.)

ASTM INTERNATIONAL

ASTM B61	-	Standard Specification for Steam or Valve Bronze Castings
ASTM B271/B271M	-	Standard Specification for Copper-Base Alloy Centrifugal Castings
ASTM B505/B505M	-	Standard Specification for Copper Alloy Continuous Castings
ASTM B584	-	Standard Specification for Copper Alloy Sand Castings for General Applications
ASTM SI 10	-	American National Standard for Metric Practice

(Copies of these documents are available online at www.astm.org.)

MANUFACTURERS STANDARDIZATION SOCIETY (MSS)

MSS SP-9 - Spot Facing for Bronze, Iron and Steel Flanges

(Copies of this document are available online at www.mss-hq.org.)

SAE INTERNATIONAL

SAE J1086 - Numbering Metals and Alloys

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Unless otherwise specified (see 6.2), the cast flanges shall be fabricated from material conforming to ASTM B61 (Unified Numbering System [UNS] alloy C92200). Material conforming to alloys C92200 or C90300 of ASTM B271/B271M, ASTM B505/B505M, or ASTM B584 is an acceptable substitute for material conforming to ASTM B61 (alloy C92200). UNS alloy designations are in accordance with SAE J1086.

3.2 Recycled/recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recycled/recovered materials to the maximum extent practicable without jeopardizing the intended use. The term “recycled/recovered materials” means materials that have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

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3.3 Type PR, with groove for preinserted rings of silver brazing alloy. Type PR flanges are grooved for preinserted rings of silver brazing alloy. The insert rings shall not be installed in the flange groove. PR flanges may be supplied without the silver brazing rings, in which case they shall be ordered separately or with the rings separately packaged and furnished with the parent component for installation in the groove prior to brazing (see 6.2). The preinserted silver brazing rings shall be of a design to fill the groove shown on figure 1, the minimum internal diameter of which shall be not less than the bore of the hub shown as diameter "S" on figure 2. Unless otherwise specified (see 6.2), the silver brazing alloy used shall conform to BCuP-5 of AWS A5.8M/A5.8. Machining details shall be as specified in table XVI and table XVII. Dimensions noted on figure 1 and figure 2 shall be as specified in tables VI through XV.

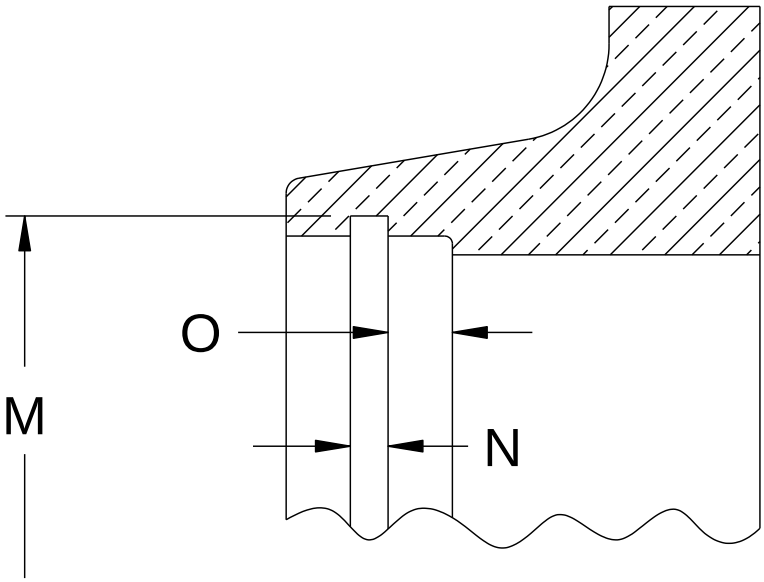


FIGURE 1. Detail dimensions of groove for preinserted ring of silver brazing alloy.

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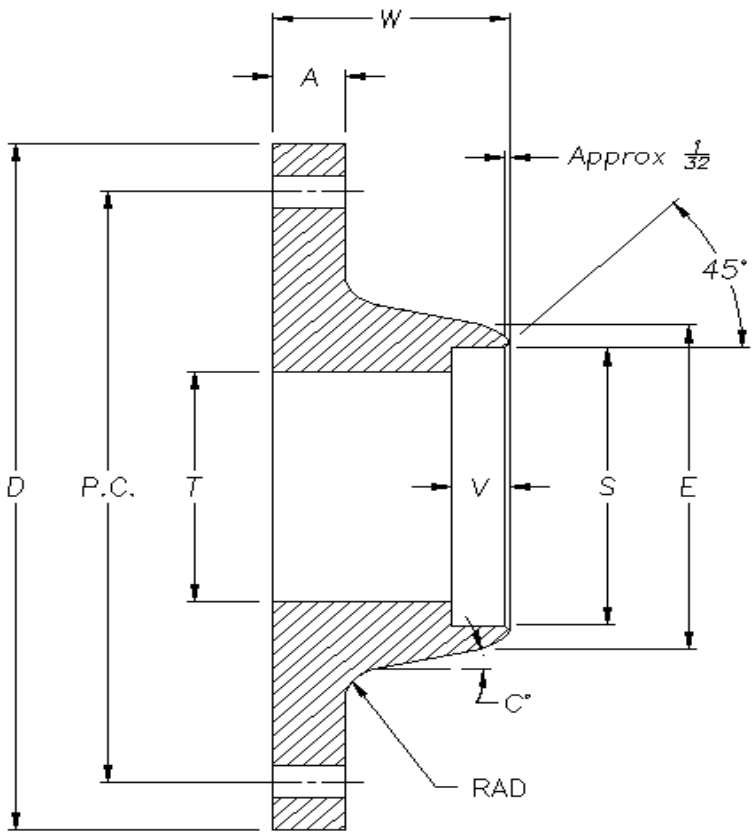


FIGURE 2. Detail dimensions of plain flanges.

3.4 Type EF, without preinserted rings (end or face feed type). Type EF flanges shall be chamfered at the outer end of the socket a distance of 0.031 inch and at an angle of about 45 degrees with the axis at the socket.

3.5 Machining.

3.5.1 Flange finish. The machined surface finish of gasket mating surfaces on the flanges shall be in accordance with ASME B46.1 and as follows:

a. For flanges of a nominal size of 12 inches or less, a finish with a circular lay (concentric) of 500 to 1000 or (spiral) 125 to 250 roughness average (R_a) produced by machining 30 to 80 serrations of uniform depth per inch of flange face width.

b. For flanges over a nominal size of 12 inches, the requirements shall be the same except that 21 to 80 serrations per inch of flange face width may be used.

Unless otherwise specified (see 6.2), flanges shall have the bore of the hub and the waterway (diameters “S” and “T” as shown on figure 2) machined with a finish of 125 or better. Flanges need not be finished on the rim, provided they are cast smooth and true. The remaining external surfaces shall not be finished except for bolt hole spot facing.

3.5.2 Flange drilling. Unless otherwise specified (see 6.2), flanges shall be drilled. Flanges shall be spot faced in accordance with MSS SP-9. The number and size of holes shall conform to tables XII through XV according to the classification of the flanges furnished.

3.5.3 Metric conversion. When metric conversion is required for machining purposes, dimensional conversion shall be in accordance with ASTM SI 10.

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3.6 Dimensions.

3.6.1 Socket. In measuring the depth of the socket in all the applicable tables specified herein, the chamfer shall be included.

3.6.2 Flange. Flange dimensions shall be as specified in the applicable figures and tables.

3.6.3 Tolerances. Unless otherwise specified herein or as shown in table III, tolerances shall be ±0.010 inch for decimal dimensions.

TABLE III. Tolerances.

NPS	D	O	S	W
	Inch	Inch	Inch	Inch
1¼ and under	+0.125 -0.000	+0.000 -0.031	+0.003 -0.000	±0.063
1½ to 2	+0.125 -0.000	+0.000 -0.031	+0.005 -0.000	±0.063
2½ to 12	+0.188 -0.000	+0.000 -0.047	+0.007 -0.000	±0.063
14 and over	+0.250 -0.000	+0.000 -0.047	+0.010 -0.000	±0.063

3.7 Marking. The size, pressure rating (“50”, “100”, “150”, “250”, or “400”), and “WOG” (Water, Oil, Gas) shall be legibly and permanently marked on the rim or back face of all flanges for identification. The rating shall be in accordance with ASME B16.5 and ASME B16.34.

3.8 Workmanship. Flanges shall be sound, smoothly cored, true to form, uniform in texture, and free of adhering sand, hard spots, cold shuts, porosity, or any other defects that may affect serviceability. They shall be thoroughly cleaned, inside and outside, and all fins and roughness that are not well rounded shall be blended.

3.9 Integrity. Flange integrity shall be demonstrated through hydrostatic, air pressure, and radiographic non-destructive tests (NDT), as applicable, as specified in 4.3 through 4.3.3.

4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall consist of the examination and tests specified in 4.2 and 4.3 as shown in table IV.

TABLE IV. Examination and tests.

Inspection	Requirement	Test method
Visual and dimensional examination	3.1 through 3.8	4.2
NDT integrity – Hydrostatic	3.9	4.3.1
NDT integrity – Air pressure	3.9	4.3.2
NDT integrity – Radiographic	3.9	4.3.3

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4.1.1 Lot. Flanges of the same type, class, size, and rating presented at one time shall be considered a lot for purposes of sampling and inspection.

4.1.2 Sampling.

4.1.2.1 Examinations and hydrostatic tests. A random sample of flanges shall be selected from each lot offered to the government in accordance with ANSI/ASQ Z1.4 at inspection level II and the acceptance quality limit (AQL) specified (see 6.2). Examination of 4.2 and the hydrostatic test of 4.3.1 shall be conducted upon these samples.

4.1.2.2 Air pressure leakage test. Every flange offered for delivery shall be subjected to the air pressure leakage test of 4.3.2.

4.1.2.3 Radiographic test. Each 400-pound flange, NPS 2½ and larger, shall be subjected to the radiographic test of 4.3.3.

4.2 Visual and dimensional examinations. Sample flanges selected as specified in 4.1.2.1 shall be visually and dimensionally examined to verify compliance with the requirements of this specification. If one or more defects are found in the sample, the entire lot represented by the sample should be rejected. If the lot is rejected, the contractor has the option of screening 100 percent of the lot for the defective characteristic(s) or providing a new lot that shall be tested as specified in table IV. Examination shall be conducted for the defects listed in table V.

TABLE V. Classification of defects.

Categories	Defects
Critical	None defined.
Major	
101	Type, class, size, or rating not as specified.
102	Material not as specified.
103	Flanges not sound, smoothly corded, true to form, uniform in texture, or not free from cold shuts and porosity.
104	Flange surfaces (internal and external) not thoroughly cleaned or fins and roughness not removed.
105	Brazing ring (if specified) missing or not as specified.
106	Socket not smooth.
107	Socket dimensions not as specified.
108	Flange dimensions not within tolerance specified.
Minor	
201	Ends (other than preinserted ring type) not chamfered as specified.
202	Marking not cast or stamped, marking incorrect, or marking illegible.

4.3 Flange integrity tests. Flange integrity (see 3.9) hydrostatic pressure leakage, air pressure leakage, and radiographic NDT tests shall be conducted on certain flanges as follows.

4.3.1 Hydrostatic pressure leakage test. Each sample flange selected as specified in 4.1.2.1 shall be subjected to a hydrostatic pressure one and a half times the rated pressure for 1 minute. Under the hydrostatic pressure, the fitting shall not leak water or sweat at any part of the surface. If the sample fails this criteria, the entire lot represented by the sample should be rejected.

4.3.2 Air pressure leakage test. Every flange shall be tested under water or with soapsuds at 100 pounds per square inch (lb/in^2) (nominal) air pressure (85 lb/in^2 minimum) for 5 to 10 seconds. No leakage shall be allowed. Since every flange shall be air pressure tested, only those failing the test shall be rejected.

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4.3.3 Radiographic test. Each 400-pound flange, 2½ NPS and larger, shall undergo a radiographic test in accordance with the applicable casting category in S9074-AR-GIB-010/278 and meet the acceptance criteria specified therein. Since each flange of these kinds are radiographically tested, only those failing the test shall be rejected.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The installation of these flanges is intended to be accomplished by silver brazing without exceeding a temperature of 1500 °F for use in naval ship piping systems. The current stock of bronze pipe flanges may be used until depleted; future requisitions for these flanges should refer to the current version of this specification. These fittings are intended for use with the following or their equivalent commercial tube and pipe specifications, standards, or UNS alloys:

a. Copper tube, MIL-T-24107 (UNS C10100, C10200, C10300, C10800, C12000, C12200, and C14200).
b. Copper-nickel alloy tube, MIL-T-16420 (UNS C71500 and C70600).
c. Brass pipe, ASTM B43 (UNS C2300).
d. Nickel-copper alloy tube, MIL-T-1368 (UNS N04400).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, size, and rating required (see 1.2).
c. Material composition, if other than as specified (see 3.1).
d. When brazing rings are to be furnished with the flanges (see 3.3).
e. If silver brazing rings are other than as specified (see 3.3).
f. If flange finish is other than as specified (see 3.5.1).
g. When flanges are to be undrilled (see 3.5.2).
h. AQL of 1.0 (see 4.1.2).
i. Packaging requirements (see 5.1).
j. Bore diameters when 400 lb/in^2 flanges are ordered (see table IX).

6.3 Subject term (key word) listing.

Brazing rings
Gas service
Oil service
Pipe standards
Steam service
Water service

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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TABLE VI. Plain flanges, 50-pound. [1/]

NPS	A (min)	D [2/]	E (min)	S [2/] (min)	T [2/]	V (min)	W [2/]	Radius
	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inch
14	0.750	19.125	14.813	14.188	14	1.188	1.937	0.500
16	0.750	21.188	16.938	16.212	16	1.188	1.937	0.500
18	0.875	23.250	19.000	18.238	18	1.188	2.063	0.500
20	0.875	25.813	21.063	20.262	20	1.188	2.063	0.500
22	0.875	27.875	23.250	22.292	22	1.313	2.188	0.563
24	1.000	30.000	25.313	24.316	24	1.313	2.313	0.563
26	1.000	32.562	27.375	26.342	26	1.313	2.313	0.563
28	1.000	34.688	29.500	28.366	28	1.375	2.375	0.625
30	1.125	36.813	31.625	30.392	30	1.438	2.563	0.625
32	1.125	39.000	33.750	32.420	32	1.500	2.625	0.625
34	1.125	41.000	35.750	34.446	34	1.625	2.750	0.688
36	1.250	43.875	37.875	36.470	36	1.688	2.875	0.688
38	1.250	46.125	40.000	38.496	38	1.750	3.000	0.688
40	1.250	48.125	42.000	40.520	40	1.813	3.000	0.750

NOTES:

[1/] See figure 2.

[2/] For tolerance, see 3.6.3.

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TABLE VII. Plain flanges, 150-pound, standard sizes. [1/]

Pipe Info		A (min)	C (min)	D [2/]	E (min)	S [2/, 3/]	T [2/, 3/]	V (min)	W [2/]	Radius	P
NPS	O.D.										
	Inches	Inch	Degree	Inches	Inches	Inches	Inches	Inches	Inches	Inch	Inch
¼ [4/]	0.540	0.375	---	3.250	0.813	0.540	0.40	0.266	0.875	0.250	0.008
⅜	0.675	0.375	---	3.375	1.000	0.675	0.54	0.313	0.938	0.250	0.008
½	0.840	0.375	---	3.563	1.156	0.840	0.71	0.375	1.000	0.250	0.008
¾	1.050	0.438	---	3.813	1.344	1.050	0.92	0.406	1.063	0.250	0.008
1	1.315	0.438	---	4.250	1.625	1.315	1.18	0.438	1.125	0.250	0.016
1¼	1.660	0.438	---	4.500	2.000	1.660	1.53	0.500	1.250	0.313	0.016
1½	1.900	0.438	---	5.063	2.250	1.900	1.77	0.625	1.375	0.313	0.016
2	2.375	0.438	---	5.563	2.750	2.375	2.24	0.656	1.375	0.313	0.016
2½	2.875	0.500	---	6.125	3.250	2.875	2.71	0.625	1.375	0.313	0.031
3	3.500	0.500	---	6.625	3.875	3.500	3.31	0.625	1.375	0.313	0.031
3½	4.000	0.500	---	7.188	4.375	4.000	3.81	0.688	1.438	0.375	0.031
4	4.500	0.500	---	7.688	4.938	4.500	4.28	0.688	1.438	0.375	0.031
4½	5.000	0.500	---	8.188	5.438	5.000	4.76	0.688	1.500	0.375	0.031
5	5.563	0.563	2	9.063	6.000	5.563	5.32	0.688	1.563	0.375	0.047
5½	6.125	0.563	2	9.563	6.500	6.125	5.88	0.750	1.688	0.438	0.047
6	6.625	0.563	2	10.125	7.063	6.625	6.36	0.750	1.688	0.438	0.047
7	7.625	0.563	2	11.313	8.063	7.625	7.36	0.813	1.813	0.438	0.047
8	8.625	0.625	2	12.375	9.125	8.625	8.33	0.938	2.000	0.438	0.063
9	9.625	0.688	2	13.938	10.188	9.625	9.25	1.000	2.188	0.500	0.063
10	10.750	0.688	2	15.000	11.313	10.750	10.38	1.063	2.250	0.500	0.063
12	12.750	0.750	2	17.625	13.438	12.750	12.25	1.250	2.500	0.500	0.063

NOTES:

1/ See figure 2.

2/ For tolerance, see 3.6.3.

3/ The eccentricity of bores “S” and “T” shall be not greater than that shown in column P where measured.

4/ For flanged gauge connections with 0.250-inch outside diameter (O.D.) or 0.500-inch O.D. tube, “S” becomes 0.250+0.003/-0.000 or 0.500+0.003/-0.000, respectively; “T” becomes 0.12 or 0.38, respectively.

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TABLE VIII. Plain flanges, 250-pound, standard sizes. [1/]

Pipe info		A (min)	C (min)	D [2/]	E (min)	S [2/, 3/]	T [2/, 3/]	V (min)	W [2/]	Radius	P
NPS	O.D.										
	Inches	Inches	Degree	Inches	Inches	Inches	Inches	Inches	Inches	Inch	Inch
¼	0.540	0.688	---	3.250	0.813	0.540	0.398	0.266	1.188	0.250	0.008
⅜	0.675	0.688	---	3.375	1.000	0.675	0.532	0.313	1.250	0.250	0.008
½	0.840	0.688	---	3.563	1.156	0.840	0.697	0.375	1.313	0.250	0.008
¾	1.050	0.688	---	3.813	1.344	1.050	0.907	0.406	1.313	0.250	0.008
1	1.315	0.750	---	4.250	1.625	1.315	1.171	0.438	1.438	0.250	0.016
1¼	1.660	0.813	---	4.500	2.000	1.660	1.502	0.500	1.625	0.313	0.016
1½	1.900	0.813	---	5.063	2.250	1.900	1.742	0.625	1.750	0.313	0.016
2	2.375	0.813	---	5.563	2.750	2.375	2.186	0.656	1.750	0.313	0.016
2½	2.875	0.938	---	6.125	3.250	2.875	2.686	0.625	1.813	0.313	0.031
3	3.500	0.938	---	6.625	3.875	3.500	3.286	0.625	1.813	0.313	0.031
3½	4.000	1.000	---	7.188	4.375	4.000	3.786	0.688	1.938	0.375	0.031
4	4.500	1.000	---	7.688	4.938	4.500	4.252	0.688	1.938	0.375	0.031
5	5.563	1.063	2	9.063	6.000	5.563	5.278	0.688	2.063	0.375	0.047
6	6.625	1.188	2	10.125	7.063	6.625	6.321	0.750	2.313	0.438	0.047
8	8.625	1.313	2	12.375	9.125	8.625	8.286	0.938	2.688	0.438	0.063
10	10.750	1.438	2	15.000	11.313	10.750	10.325	1.063	3.000	0.500	0.063
12	12.750	1.500	2	17.625	13.438	12.750	12.322	1.250	3.250	0.500	0.063

NOTES:

[1/] See figure 2.

[2/] For tolerance, see 3.6.3.

[3/] The eccentricity of bores “S” and “T” shall be not greater than that shown in column P where measured.

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TABLE IX. Plain flanges, 400-pound, standard sizes. [1/]

Pipe info		A (min)	C (min)	D [2/]	E (min)	S [2/, 3/]	T [2/, 3/]	V (min)	W [2/]	Radius	P
NPS	O.D.										
	Inches	Inches	Degree	Inches	Inches	Inches	Inches	Inches	Inches	Inch	Inch
¼ [4/]	0.540	0.688	---	3.750	0.875	0.540	0.41	0.266	1.188	0.250	0.008
⅜	0.675	0.688	---	3.875	1.063	0.675	0.54	0.313	1.250	0.250	0.008
½	0.840	0.688	---	4.000	1.250	0.840	0.71	0.375	1.313	0.250	0.008
¾	1.050	0.688	---	4.313	1.438	1.050	0.92	0.406	1.313	0.250	0.008
1	1.315	0.750	---	5.063	1.750	1.315	1.18	0.438	1.438	0.250	0.016
1¼	1.660	0.813	---	5.375	2.125	1.660	1.53	0.500	1.625	0.313	0.016
1½	1.900	0.813	---	5.938	2.375	1.900	1.77	0.625	1.750	0.313	0.016
2	2.375	0.813	---	6.500	2.875	2.375	2.24	0.656	1.750	0.313	0.016
2½	2.875	0.938	---	7.563	3.375	2.875	[5/]	1.000	2.313	0.500	0.031
3	3.500	0.938	---	8.125	4.000	3.500	[5/]	1.000	2.375	0.500	0.031
3½	4.000	1.000	---	8.688	4.500	4.000	[5/]	1.000	2.438	0.500	0.031
4	4.500	1.000	---	9.250	5.000	4.500	[5/]	1.125	2.625	0.563	0.031
4½	5.000	1.000	---	9.813	5.625	5.000	[5/]	1.125	2.625	0.563	0.031
5	5.563	1.063	2	10.375	6.188	5.563	[5/]	1.250	2.813	0.563	0.047
5½	6.125	1.125	2	11.375	6.750	6.125	[5/]	1.250	2.938	0.625	0.047
6	6.625	1.188	2	11.938	7.313	6.625	[5/]	1.375	3.125	0.625	0.047
7	7.625	1.250	2	13.125	8.375	7.625	[5/]	1.375	3.250	0.688	0.047
8	8.625	1.313	2	14.750	9.375	8.625	[5/]	1.500	3.500	0.688	0.063
9	9.625	1.375	2	15.875	10.500	9.625	[5/]	1.625	3.750	0.750	0.063
10	10.750	1.438	2	17.000	11.625	10.750	[5/]	1.750	4.000	0.750	0.063

NOTES:

[1/] See figure 2.

[2/] For tolerance, see 3.6.3.

[3/] The eccentricity of bores “S” and “T” shall be not greater than that shown in column P where measured.

[4/] For flanged gauge connections with 0.250-inch O.D. or 0.500-inch O.D. tube, “S” becomes 0.250+0.003/-0.000 or 0.500+0.003/-0.000, respectively; “T” becomes 0.12 or 0.38, respectively.

[5/] Bore to suit inside diameter of tube (see 6.2).

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TABLE X. Machining details for type PR, 50-pound flanges, standard sizes. [1/]

NPS	O [2/]	N		M (diameter)	
		Min	Max	Min	Max
	Inch	Inch	Inch	Inches	Inches
14	0.395	0.398	0.408	14.366	14.386
16	0.395	0.398	0.408	16.390	16.410
18	0.395	0.398	0.408	18.416	18.436
20	0.395	0.398	0.408	20.440	20.460
22	0.457	0.398	0.408	22.506	22.526
24	0.457	0.398	0.408	24.530	24.550
26	0.457	0.398	0.408	26.556	26.576
28	0.488	0.398	0.408	28.580	28.600
30	0.520	0.398	0.408	30.606	30.626
32	0.551	0.398	0.408	32.634	32.654
34	0.613	0.398	0.408	34.660	34.680
36	0.645	0.398	0.408	36.684	36.704
38	0.676	0.398	0.408	38.710	38.730
40	0.707	0.398	0.408	40.734	40.754

NOTES:

[1/] See figure 1.

[2/] For tolerance, see 3.6.3.

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TABLE XI. Machining details for type PR, 100-, 150-, 250-, and 400-pound flanges, standard sizes. [1/]

Pipe info		O [2/]	N		M (diameter)	
NPS	O.D.		Min	Max	Min	Max
	Inches	Inch	Inch	Inch	Inches	Inches
¼	0.540	0.100	0.068	0.078	0.619	0.639
⅜	0.675	0.117	0.083	0.093	0.764	0.784
½	0.840	0.149	0.083	0.093	0.929	0.949
¾	1.050	0.154	0.099	0.109	1.159	1.179
1	1.315	0.154	0.130	0.140	1.424	1.444
1¼	1.660	0.185	0.130	0.140	1.769	1.789
1½	1.900	0.217	0.192	0.202	2.011	2.031
2	2.375	0.232	0.192	0.202	2.486	2.506
2½	2.875	0.217	0.192	0.202	2.988	3.008
3	3.500	0.185	0.255	0.265	3.613	3.633
3½	4.000	0.217	0.255	0.265	4.113	4.133
4	4.500	0.217	0.255	0.265	4.673	4.693
4½ [3/]	5.000	0.217	0.255	0.265	5.173	5.193
5	5.563	0.217	0.255	0.265	5.736	5.756
5½ [3/]	6.125	0.248	0.255	0.265	6.298	6.318
6	6.625	0.248	0.255	0.265	6.798	6.818
7 [3/]	7.625	0.247	0.320	0.330	7.798	7.818
8	8.625	0.270	0.398	0.408	8.798	8.818
9 [3/]	9.625	0.301	0.398	0.408	9.798	9.818
10	10.750	0.333	0.398	0.408	10.923	10.943
11 [3/, 4/]	11.750	0.364	0.398	0.408	11.963	11.983
12 [4/]	12.750	0.426	0.398	0.408	12.963	12.983

NOTES:

[1/] See figure 1.

[2/] For tolerance, see 3.6.3.

[3/] Not for 250-pound flange.

[4/] Not for 400-pound flange.

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TABLE XII. Drilling dimensions, 50-pound flanges. [1/]

NPS	Flange joint			
	Number of holes	Diameter of holes	Pitch circle	Pitch cord
		Inches	Inches	Inches
14	19	0.938	17.375	2.86
16	20	0.938	19.438	3.04
18	22	0.938	21.500	3.06
20	24	1.063	23.813	3.11
22	26	1.063	25.875	3.12
24	28	1.063	28.000	3.13
26	30	1.188	30.313	3.17
28	32	1.188	32.438	3.18
30	35	1.188	34.563	3.10
32	36	1.188	36.750	3.20
34	36	1.188	38.750	3.38
36	36	1.313	41.375	3.61
38	36	1.313	43.625	3.80
40	36	1.313	45.625	3.98

NOTE:

[1/] See figure 2.

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TABLE XIII. Drilling dimensions, 150-pound flanges, standard sizes. [1/]

Pipe info		Flange joint			
NPS	O.D.	Number of holes	Diameter of holes	Pitch circle	Pitch cord
	Inches		Inch	Inches	Inches
¼	0.540	3	0.563	2.13	1.84
⅜	0.675	3	0.563	2.25	1.95
½	0.840	3	0.563	2.44	2.11
¾	1.050	4	0.563	2.69	1.90
1	1.315	4	0.563	3.13	2.21
1¼	1.660	4	0.563	3.38	2.39
1½	1.900	6	0.563	3.94	1.97
2	2.375	6	0.563	4.44	2.22
2½	2.875	6	0.563	5.00	2.50
3	3.500	8	0.563	5.50	2.10
3½	4.000	8	0.563	6.06	2.32
4	4.500	8	0.563	6.56	2.51
4½	5.000	10	0.563	7.06	2.18
5	5.563	10	0.688	7.81	2.41
5½	6.125	10	0.688	8.31	2.57
6	6.625	12	0.688	8.88	2.30
7	7.625	12	0.688	10.00	2.59
8	8.625	14	0.688	11.06	2.46
9	9.625	14	0.813	12.38	2.75
10	10.750	15	0.813	13.44	2.79
11	11.750	16	0.813	15.00	2.93
12	12.750	18	0.813	16.06	2.79

NOTE:

[1/] See figure 2.

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TABLE XIV. Drilling dimensions, 250-pound flanges, standard sizes. [1/]

NPS	O.D.	Flange joint		
		Number of holes	Diameter of holes	Pitch circle
	Inches		Inch	Inches
¼	0.540	3	0.563	2.13
⅜	0.675	3	0.563	2.25
½	0.840	3	0.563	2.44
¾	1.050	4	0.563	2.69
1	1.315	4	0.563	3.13
1¼	1.660	4	0.563	3.38
1½	1.900	6	0.563	3.94
2	2.375	6	0.688	4.44
2½	2.875	6	0.688	5.00
3	3.500	8	0.688	5.50
3½	4.000	8	0.688	6.06
4	4.500	8	0.688	6.56
5	5.563	10	0.688	7.81
6	6.625	12	0.688	8.88
8	8.625	14	0.688	11.06
10	10.750	15	0.813	13.44
12	12.750	18	0.813	16.06

NOTE:

[1/] See figure 2.

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TABLE XV. Drilling dimensions, 400-pound flanges, standard sizes. [1/]

Pipe info		Flange joint			
NPS	O.D.	Number of holes	Diameter of holes	Pitch circle	Pitch cord
	Inches		Inches	Inches	Inches
¼	0.540	3	0.563	2.63	2.27
⅜	0.675	4	0.563	2.75	1.94
½	0.840	4	0.563	2.88	2.03
¾	1.050	4	0.563	3.19	2.25
1	1.315	5	0.688	3.75	2.20
1¼	1.660	5	0.688	4.06	2.39
1½	1.900	6	0.688	4.63	2.31
2	2.375	7	0.688	5.19	2.25
2½	2.875	8	0.813	6.00	2.30
3	3.500	8	0.813	6.56	2.51
3½	4.000	9	0.813	7.13	2.44
4	4.500	9	0.813	7.69	2.63
4½	5.000	10	0.813	8.25	2.55
5	5.563	11	0.813	8.81	2.48
5½	6.125	11	0.938	9.63	2.71
6	6.625	12	0.938	10.19	2.64
7	7.625	12	0.938	11.38	2.94
8	8.625	13	1.063	12.75	3.05
9	9.625	14	1.063	13.88	3.09
10	10.750	15	1.063	15.00	3.12

NOTE:

[1/] See figure 2.

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TABLE XVI. Dimensions of silver brazing ring for 50-pound flanges. 1/

NPS	Q±0.003	K±0.003	H	F	
	Ring thickness	Ring width	O.D.	Min	Max
	Inch	Inch	Inches	Inch	Inch
14	0.096	0.391	14.386	0.073	0.266
16	0.096	0.391	16.410	0.073	0.266
18	0.096	0.391	18.436	0.073	0.266
20	0.096	0.391	20.460	0.073	0.266
22	0.115	0.391	22.526	0.073	0.302
24	0.115	0.391	24.550	0.073	0.302
26	0.115	0.391	26.578	0.073	0.302
28	0.115	0.391	28.600	0.073	0.302
30	0.115	0.391	30.626	0.073	0.302
32	0.115	0.391	32.654	0.073	0.302
34	0.115	0.391	34.680	0.073	0.302
36	0.115	0.391	36.704	0.073	0.302
38	0.115	0.391	38.730	0.073	0.302
40	0.115	0.391	40.754	0.073	0.302

NOTE:
1/ See figure 3.

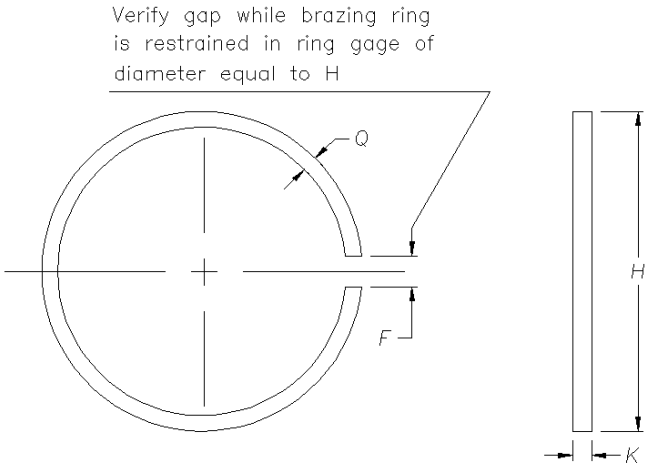


FIGURE 3. Preinserted braze ring detail.

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TABLE XVII. Dimensions of silver brazing ring for 100-, 150-, 250-, and 400-pound flanges. [1/]

NPS	Q±0.003	K±0.003	H	F	
	Ring thickness	Ring width	O.D.	Min	Max
	Inch	Inch	Inches	Inch	Inch
¼	0.035	0.063	0.639	0.073	0.130
⅜	0.040	0.078	0.784	0.073	0.130
½	0.040	0.078	0.949	0.073	0.130
¾	0.050	0.094	1.179	0.073	0.130
1	0.050	0.125	1.444	0.073	0.135
1¼	0.050	0.125	1.789	0.073	0.135
1½	0.050	0.188	2.031	0.073	0.135
2	0.050	0.188	2.506	0.073	0.135
2½	0.050	0.188	3.008	0.073	0.135
3	0.050	0.250	3.633	0.073	0.167
3½	0.050	0.250	4.133	0.073	0.167
4	0.080	0.250	4.693	0.073	0.229
5	0.080	0.250	5.756	0.073	0.229
6	0.080	0.250	6.818	0.073	0.261
7	0.080	0.391	7.818	0.073	0.261
8	0.080	0.391	8.818	0.073	0.261
9	0.080	0.391	9.818	0.073	0.261
10	0.080	0.391	10.943	0.073	0.261
12	0.096	0.391	12.983	0.073	0.261

NOTE:

[1/] See figure 3.

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CONCLUDING MATERIAL

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Navy – SH
Air Force – 71

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Navy − SH
(Project 4730-2020-072)

Review activities:
Navy – OS, YD
DLA – CC

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